



MINING DEALER BEST PRACTICE

C15 & C18 Engines Camshaft Bearings Assembly/ Disassembly Stand

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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March 2015
1502-4.03-1370

1.0 Introduction

Borusan Makina Kazakhstan has designed the tool to easily and safely assemble/disassemble the bearings of the C15 & C18 engines camshaft with one technician.

Before creating this tool, it involved two technicians and took about 40 minutes to assemble the bearings of the camshaft. It was inconvenient and required more effort and man-hours, mostly because a manual tool was used and it had to measure the bearings one by one so that the oil grooves would match during assembling.

Thus, this new tool or stand enables to safely and simultaneously assemble/ disassemble the bearings of the camshaft and reduce the efforts spent and optimize the man hours allocated.

2.0 Best Practice Description

This stand consists of the table, 7 x pressing shafts, 2 x studs, 6 x cylinder head mounting bolts (Part No. 6I4025), 1 x retainer, hydraulic press and additional adjusting fixture (Part No. 4N4941 – Adjustment Rod (1 pc) & 6B6684 – Nut (2 pcs)).



The Table of the stand – the main part of the stand

3.0 Implementation Steps

The stand in use (the Bearings Assembly Process).

1. Prepare the hydraulic press for use.
2. Prepare the tools required to assemble the bearings.

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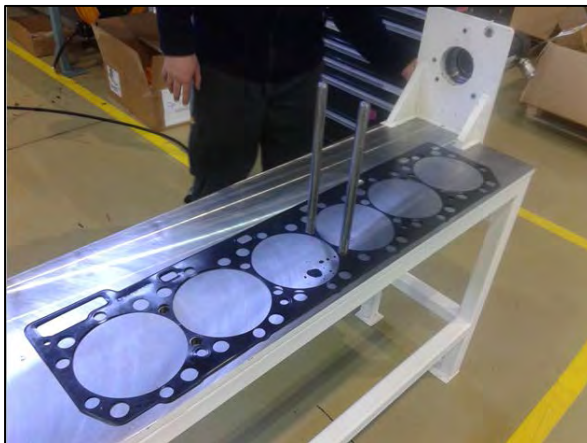
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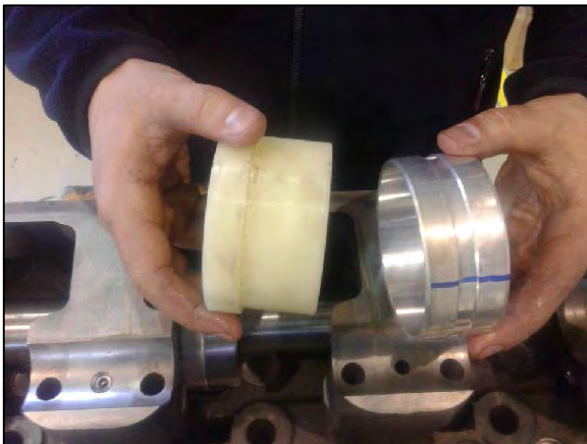
3. Put the marks on the bearings.

4. Install the studs in the stand, and then put the gasket. Lower the cylinder head onto the stand by the crane and tighten the mounting bolts with the torque of 150 LBFT.

EDITOR'S NOTE: Teflon (or high density plastic) could be considered instead of the head gasket.



5. Install the bearings on the shaft one by one.



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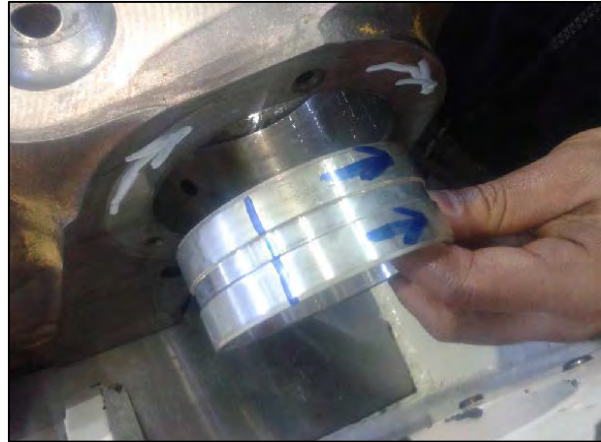
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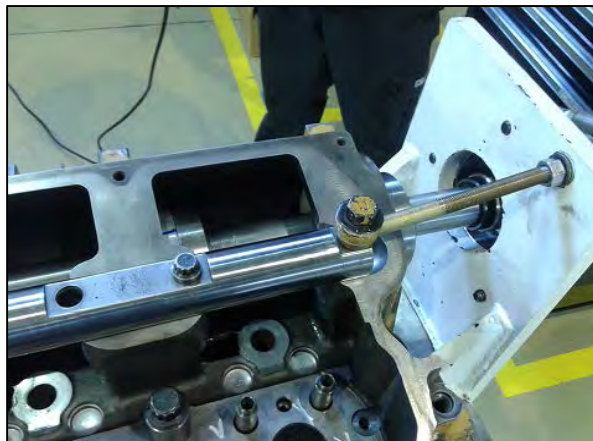
6. It is necessary that the locks of the bearings should match the turning direction of the camshaft.



7. Lubricate the bearings with oil.
8. Install the retainer and connect the hydraulic press.



9. Install the additional adjusting fixture in order to prevent the metal from rolling and then apply the pressure from the hydraulic press.



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10. Remove the retainer from the 7th bearing and assure that the bearings have reached the depth required.



11. Remove the pressing shaft and then remove the cylinder head from the stand.

This stand can also be used to disassemble bearings, simply reverse sequence of work. For moving the stand from the assemble area to the disassemble area, equip the stand with wheels.

4.0 Benefits

- Easy and safe bearings assembly/ disassembly processes.
- Reduced efforts spent by the technician
- Optimized man hours for these processes.

5.0 Resources Required

- Machine Shop and materials to fabricate the stand (see the drawings attached).

6.0 Supporting Attachments / References

- Specifications-Model.pdf
- St 01.000_SB-Model.pdf
- St 01.001-Model.pdf
- St 01.002-Model.pdf
- St 01.003-Model.pdf

7.0 Related Best Practices

0408-4.4-1119 - 3500 Engine Series Camshaft Bearing Installation Tool

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